

OMAR FADLALLAH

AI Engineer | LLM Systems | RAG Architectures | NLP

Mansoura, Egypt • +20 127 745 7703 • omaryaser311@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer who started in Data Analysis and BI before moving deliberately into ML and production GenAI — mastering the data foundation first, then scaling into agentic systems. Built enterprise-grade RAG pipelines achieving **sub-50ms P50 latency**, a multi-agent AI Operating System deployed on Hugging Face Spaces, and a **70% CER reduction** on Arabic legal OCR through fine-tuning a 4B vision-language model. Interned at Devoteam delivering enterprise AI PoCs; **NVIDIA DLI-certified** in Applied Deep Learning and Gen AI.

TECHNICAL SKILLS

LLM & RAG: Retrieval-Augmented Generation, Prompt Engineering, LLM Fine-tuning (LoRA / QLoRA), Agentic Systems, Hybrid Retrieval, Cross-encoder Re-ranking, Multi-step Reasoning

Frameworks: LangChain, LangGraph, HuggingFace Transformers, PyTorch, FastAPI, Gradio

Vector & Search: Qdrant, FAISS, BM25, Semantic Search, Redis (semantic cache)

Evaluation & MLOps: MLflow, Prometheus, Grafana, Celery, ZenML, Docker

Testing: Unit Testing (pytest), API Testing, Load Testing (Locust / k6)

Cloud, DevOps & Serving: AWS, Hugging Face Spaces, GitHub Actions, vLLM, ONNX Runtime

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy, SQLAlchemy

Languages: Python (primary), SQL, C++, R • Arabic (native), English (fluent)

Certifications: NVIDIA DLI – Applied Deep Learning • NVIDIA DLI – Generative AI • AWS Practical Training (DataCamp)

PROFESSIONAL EXPERIENCE

Analytics & Agentic AI Intern • [Kayfa](#)

May 2026 – Jun 2026

- Engineered and deployed a Hybrid RAG Course Advisor Agent (FastAPI, PydanticAI, LlamaIndex, MongoDB, Groq) with semantic retrieval, streaming responses, and JWT-secured conversation management — live on Hugging Face Spaces. [Live Demo](#)
- Built a Student Analytics Dashboard (Streamlit, MongoDB Atlas, Plotly) aggregating 500+ students from 8 data sources, with a 41-point ETL audit and a K-Means risk-scoring model for 4-tier student segmentation. [Live Demo](#)
- Automated CI/CD pipelines (GitHub Actions) for multi-container, zero-downtime deployments across Hugging Face Spaces.

AI Engineer Intern • [Devteam](#)

Dec 2025 – Feb 2026

- Built a real-time driver drowsiness detection system (OpenCV, Keras/TensorFlow, ONNX) using Haar Cascade face/eye detection and a CNN eye-state classifier, converting the model to ONNX for faster cross-platform inference.
- Built an end-to-end marketing campaign optimization pipeline (ZenML, MLflow, FastAPI) on 200,000+ campaign records across 6 channels and 5 audience segments, training and serving a conversion-rate prediction model via REST API.
- Collaborated with product and data teams across 2 enterprise AI PoCs, translating client requirements into scoped deliverables.

KEY PROJECTS

AI Platform for Computational Drug Discovery & Repurposing — 5 Independent Production Services ★ *Graduation Project* — **Lead AI & Backend Engineer • Team of 9.** | [Desktop App](#) | [Backend Repo](#)

Flask • FAISS-IVF • RDKit • Chemprop • PyTorch Lightning • DeepPurpose • OpenMM • Docker • asyncio • Celery • Redis

- Platform Architecture:** Designed and led development of 5 AI services (Chemical RAG, ADMET Prediction, Drug Repurposing, Molecular Dynamics, Scientific OS), each maintained in its own repository with a dedicated CI pipeline gating deployment to Hugging Face Spaces — unified through a central backend that owns cross-service orchestration and integration.
- Chemical RAG:** Indexed 1M+ compounds in a FAISS-IVF vector store with sub-100ms similarity retrieval; grounded Llama-3.1-8B with RDKit-extracted molecular properties for reliable scientific reasoning without fine-tuning.
- ADMET Prediction:** Trained 5 task-specific MPNN models on molecular property data; engineered a concurrent async inference engine with parallel execution — cutting single-molecule prediction to 50–150ms on CPU with no GPU dependency.
- Drug Repurposing:** High-throughput screening of 5,000+ approved drugs against protein targets using binding-affinity models.
- Molecular Dynamics:** Productized a multi-step simulation protocol into a single autonomous API, offloading long-running jobs to Celery/Redis workers; deployed via multi-stage Docker with GitHub Actions CI/CD.
- Domain Validation:** Recruited 7 medical and pharmacy faculty members as actual end-users, conducting structured usability testing and incorporating their feedback on clinical relevance and interface usability into the final release.
- Multi-Agent Orchestration:** Architected a multi-agent drug discovery backend (Groq LLM orchestrator, 3 specialist agents: Chemical, Medical, Conversational) with a hybrid RAG pipeline (LlamaIndex, multilingual Arabic/English embeddings), a real-time bilingual voice agent (WebSocket, VAD, Whisper STT, OpenAI TTS), and production deployment on HuggingFace Spaces. [Live Demo](#)

Atlas AI Platform — Enterprise Multi-Tenant RAG — [GitHub](#) | [Demo](#) | [LangChain](#) · [LangGraph](#) · [Qdrant](#) · [Redis](#) · [Prometheus](#) · [Grafana](#)

- Designed a multi-tenant SaaS RAG platform with JWT authentication and strict per-tenant data isolation, enabling secure document Q&A for multiple organizations on a shared infrastructure.
- Built hybrid retrieval pipeline combining dense vector search (Qdrant), BM25 keyword matching, and cross-encoder re-ranking — improving answer relevance and reducing hallucinations compared to single-retriever baselines.
- Implemented a 3-tier caching architecture (in-memory → Redis semantic cache → database) achieving sub-50ms P50 API latency under load and cutting LLM API call volume by 60–80% through semantic cache hit reuse.
- Fine-tuned Qwen2.5-1.5B with LoRA/QLoRA on Arabic instruction-following data using 4-bit NF4 quantization, improving domain-specific Arabic Q&A accuracy while keeping inference cost minimal on consumer hardware.
- Wrote a comprehensive test suite (pytest unit tests, API contract tests, Locust load tests) with per-request cost and token tracking; real-time observability via Prometheus and Grafana (latency, token usage, drift alerts).
- Built a LangGraph-based agentic reasoning layer supporting multi-step query decomposition and tool use for complex questions.

Arabic Legal OCR — Fine-tuning Vision-Language Models for Structured Extraction — [GitHub](#) | [PyTorch](#) · [LlamaFactory](#) ·

[LoRA/QLoRA](#) · [vLLM](#) · [Gemma-3-4b-it](#) · [HuggingFace PEFT](#)

- Fine-tuned Gemma-3-4b-it (4B params) using LoRA on Arabic legal documents, reducing Character Error Rate from 0.74 to 0.22 — a 70% CER improvement — and lifting OCR accuracy from ~26% to ~78%.
- Scraped and processed 72+ legal documents from the Saudi Ministry of Justice API; used knowledge distillation with Gemini 1.5 Pro to generate structured JSON training labels.
- Built a dual-task extraction pipeline outputting structured document content and metadata (issuing authority, dates, signatures) with confidence scoring.
- Deployed production inference via vLLM with multi-LoRA support; trained with 4-bit NF4 quantization, enabling highly efficient fine-tuning on a consumer-grade GPU with zero performance loss.

Document Intelligence Platform — Email-to-Decision Document Automation — [GitHub](#) | [Live Demo](#) | [Python](#) · [FastAPI](#) ·

[React/TypeScript](#) · [MongoDB](#) · [PaddleOCR](#) · [Groq \(gpt-oss-20b\)](#) · [Docker](#)

- Architected a layered document-intelligence pipeline that turns an incoming email and its attachments (PDF, DOCX, scanned images) into a fully typed, auditable routing decision — APPROVE / ROUTE_TO_HUMAN_REVIEW / REJECT.
- Decoupled LLM extraction from decision-making: structured data is extracted via an LLM, then validated exclusively through deterministic, code-based business rules (totals, tax, regex, vendor whitelist) — no field trusted purely on model output.
- Built a cross-document matching service that reconciles invoice line items against delivery-note line items by item code and normalized product name, flagging quantity mismatches and unmatched items automatically.
- Designed a Strategy + Factory-based parsing layer (PDF/DOCX/image) with OCR fallback for scanned documents, and a confidence-scoring validation engine computing a 0–1 score from rule pass rate.

LLMOps Platform — Production Prompt Management System — [GitHub](#) | [Demo](#) | [Python](#) · [FastAPI](#) · [React/Vite](#) · [Celery](#) · [PostgreSQL](#)

· [Docker](#) · [Redis](#) · [LangChain](#)

- Built a full-stack LLMOps platform (FastAPI + React/Vite) to prevent silent prompt regressions and untracked LLM costs in production — with prompt versioning, API key authentication, and Redis-backed rate limiting (60 req/min per key).
- Designed a regression-testing engine that runs every prompt version against a fixed set of golden examples, using an LLM-as-judge evaluator to score outputs and flag hallucinations before they reach production.
- Implemented per-run cost, token, and latency tracking alongside prompt version diffing and one-click rollback, giving full visibility into what changed and what it cost.
- Offloaded long-running LLM calls and batch evaluations to Celery/Redis workers for async, non-blocking execution; data layer uses PostgreSQL/SQLAlchemy with Alembic migrations.

EDUCATION

B.Sc. Computer Science · [Faculty of Computers and Information, Mansoura University](#)

Oct 2022 – Jul 2026

GPA: 3.7 / 4.0 · Top 5% of cohort · Graduated: July 2026 · Military Status: Exempted